Lesson: HR Analytics Dashboard — Project Plan & Requirements

Project Objective

Design and build a comprehensive HR Analytics Dashboard in Power BI to help HR managers monitor performance, retention, department KPIs, and engagement.

Dataset: Employee_Performance (from HR_Analytics.csv)

Columns included: Employment_id, Department, Age, Job Title, Hire_Date, Years_at_company, Education_level, Performance_Score, Monthly_Salary, Work_Hours_per_Week, Project_Handled, Overtime_Hours, Sick_Days, Remote_Work_Frequency, Team_Size, Training_Hours, Promotions, Employee_Satisfaction_Score, Resigned (Yes/No)

Power Query Editor — Data Preparation Steps

Rename columns, change data types (dates, numeric, text), remove duplicates on Employment_id, handle missing values (impute or remove), create calculated columns (Tenure Category, Overtime Category), create Date table using CalendarAuto, create lookup tables for Department/Education_level if needed.

Calculated Columns (DAX examples)

Tenure Category = SWITCH(TRUE(), Employee_Performance[Years_at_company] <= 2, "New", Employee_Performance[Years_at_company] <= 5, "Mid", "Veteran")
Overtime Category = IF(Employee_Performance[Overtime_Hours] > 10, "High", "Low")

Data Model & Relationships

Create a star schema: central fact table 'Employee_Performance' linked to Date table via Hire_Date (one-to-many). Use lookup tables for Department and Education_level. Avoid circular dependencies; prefer single-direction relationships.

DAX Measures (Key KPIs)

Employee Count = DISTINCTCOUNT(Employee_Performance[Employment_id]) Resignation Rate = DIVIDE(CALCULATE(COUNTROWS(Employee_Performance), Employee_Performance[Resigned] = "Yes"), [Employee Count], 0) Avg Performance = AVERAGE(Employee_Performance[Performance_Score]) Avg Salary = AVERAGE(Employee_Performance[Monthly_Salary]) Avg Training = AVERAGE(Employee_Performance[Training_Hours]) Avg Satisfaction = AVERAGE(Employee_Performance[Employee_Satisfaction_Score]) Overtime Utilization = DIVIDE(SUM(Employee_Performance[Overtime_Hours]), SUM(Employee_Performance[Work_Hours_per_Week]), 0) Sick Days per Emp = DIVIDE(SUM(Employee_Performance[Sick_Days]), [Employee Count], 0) Remote Adoption = DIVIDE(CALCULATE(COUNTROWS(Employee_Performance), Employee_Performance[Remote_Work_Frequency] = "Often"), [Employee Count], 0) Promotion Rate = DIVIDE(SUM(Employee_Performance[Promotions]), [Employee Count], 0) Avg Tenure = AVERAGE(Employee_Performance[Years_at_company])

Report Pages & Visuals

Page 1 — Executive Summary: Cards (Total Employees, Resignation Rate, Avg Performance, Avg Salary), Line chart (Resignation rate over time), Stacked column (Department-wise Satisfaction), Slicers (Department, Hire Date), KPI YoY.
Page 2 — Department Insights: Bar chart (Employees per Department), Heatmap (Avg Salary vs Performance), Pie (Education Level), KPI Avg Tenure.

Page 3 — Employee Engagement: Gauge (Satisfaction), Donut (Remote Work Frequency), Clustered column (Avg Overtime by Job Title), Scatter (Training vs Performance).
Page 4 — Retention & Promotions: Matrix (Promotions by Dept & Education), Bar (Sick Days vs Resigned), Line (Training Trend), Card (Promotion Rate).
Page 5 — Filters & Slicers: Department, Job Title, Education Level, Remote Work Frequency, Tenure Category, Resigned (Yes/No).

Power BI Features to Apply

DAX for KPIs, custom tooltips, drillthrough pages, bookmarks, conditional formatting, sync slicers, RLS, scheduled refresh and app publishing.

Advanced Features & Maintenance

Row-Level Security for managers, Power Automate alerts, Paginated reports, Version control, Monthly data quality checks.

Delivery Checklist & Suggested Timeline

Week 1: Data ingestion & cleaning. Week 2: Data model & basic measures. Week 3: Executive & Department visuals. Week 4: Engagement & Retention pages. Week 5: RLS, testing, and publishing. Deliverables: PBIX file, documentation, published app with scheduled refresh.